Stack Implementation using Array in C

Source Code

```
#include <stdio.h>
#include <stdbool.h>
#include <stdlib.h>

#define MAX_SIZE 10

int stack[MAX_SIZE];
int top = -1;

bool isEmpty();
bool isFull();

// Function to push element onto the stack
void push(){
    if(isFull()){
        printf("\nStack Overflow");
    }
    else{
        printf("\nEnter Element: ");
        scanf("%d",&stack[++top]);
        printf("\nElement pushed");
    }
}

// Function to pop top element
int pop(){
    if(isEmpty()){
        printf("\nStack Underflow");
    }
    else{
        printf("Popped item: %d\n", stack[top]);
        return stack[top--];
    }
}
```

```c
// Function to check if stack is full
bool isFull(){
    return (top == MAX_SIZE-1);
}

// Function to check if stack is empty
bool isEmpty(){
    return (top == -1);
}

// Function to display stack elements
void display(){
    if(isEmpty()){
        printf("\nStack is empty.");
    }
    else{
        printf("\n");
        for(int i = top; i >= 0; i--){
            printf("%d\n", stack[i]);
        }
    }
}

// Main function with menu
int main() {
    int choice;
    int item;

    while (1) {
        printf("\n====== STACK MENU ======\n");
        printf("1. Push\n");
        printf("2. Pop\n");
        printf("3. Check if Full\n");
        printf("4. Check if Empty\n");
        printf("5. Display\n");
        printf("6. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
```

```
            case 1:
                push();
                break;
            case 2:
                item = pop();
                break;
            case 3:
                if (isFull())
                    printf("Stack is FULL!\n");
                else
                    printf("Stack is NOT full.\n");
                break;
            case 4:
                if (isEmpty())
                    printf("Stack is EMPTY!\n");
                else
                    printf("Stack is NOT empty.\n");
                break;
            case 5:
                display();
                break;
            case 6:
                printf("Exiting...\n");
                exit(0);
            default:
                printf("Invalid choice! Try again.\n");
        }
    }
    return 0;
}
```

Output :

```
====== STACK MENU ======
1. Push
2. Pop
3. Check if Full
4. Check if Empty
5. Display
6. Exit
Enter your choice: 1

Enter Element: 10

Element pushed
====== STACK MENU ======
1. Push
2. Pop
3. Check if Full
4. Check if Empty
5. Display
6. Exit
Enter your choice: 1

Enter Element: 20

Element pushed
====== STACK MENU ======
1. Push
2. Pop
3. Check if Full
4. Check if Empty
5. Display
6. Exit
Enter your choice: 1
```

```
Enter Element: 30

Element pushed
====== STACK MENU ======
1. Push
2. Pop
3. Check if Full
4. Check if Empty
5. Display
6. Exit
Enter your choice: 5

30
20
10

====== STACK MENU ======
1. Push
2. Pop
3. Check if Full
4. Check if Empty
5. Display
6. Exit
Enter your choice: 2
Popped item: 30

====== STACK MENU ======
1. Push
2. Pop
```

```
3. Check if Full
4. Check if Empty
5. Display
6. Exit
Enter your choice: 3
Stack is NOT full.

====== STACK MENU ======
1. Push
2. Pop
3. Check if Full
4. Check if Empty
5. Display
6. Exit
Enter your choice: 4
Stack is NOT empty.

====== STACK MENU ======
1. Push
2. Pop
3. Check if Full
4. Check if Empty
5. Display
6. Exit
Enter your choice: 6
Exiting...
```